PULL-PANDA

Inspired by automated PR analysis

Group 09

IT314 – Software Engineering
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Objective

To streamline and enhance the code review process using AI, static analyzers, and RAG, enabling faster feedback, improved code quality, and an efficient GitHub-native workflow.

Milestones

Sprint 1 :
Cloud LLM MVP

Built the basic PR review pipeline: fetch PR, send diff to cloud LLM, get structured review, and show it via simple UI.

Sprint 2 :
Automation + Static Analysis + Initial Prompt Selector

Added GitHub Actions automation, integrated ESLint/Pylint/Bandit, and introduced the first ML-based prompt selector using Random Forest.

Sprint 3 :
Repository Indexing (RAG)

Implemented full repo indexing with embeddings and retrieved context files during PR review using a local RAG setup.

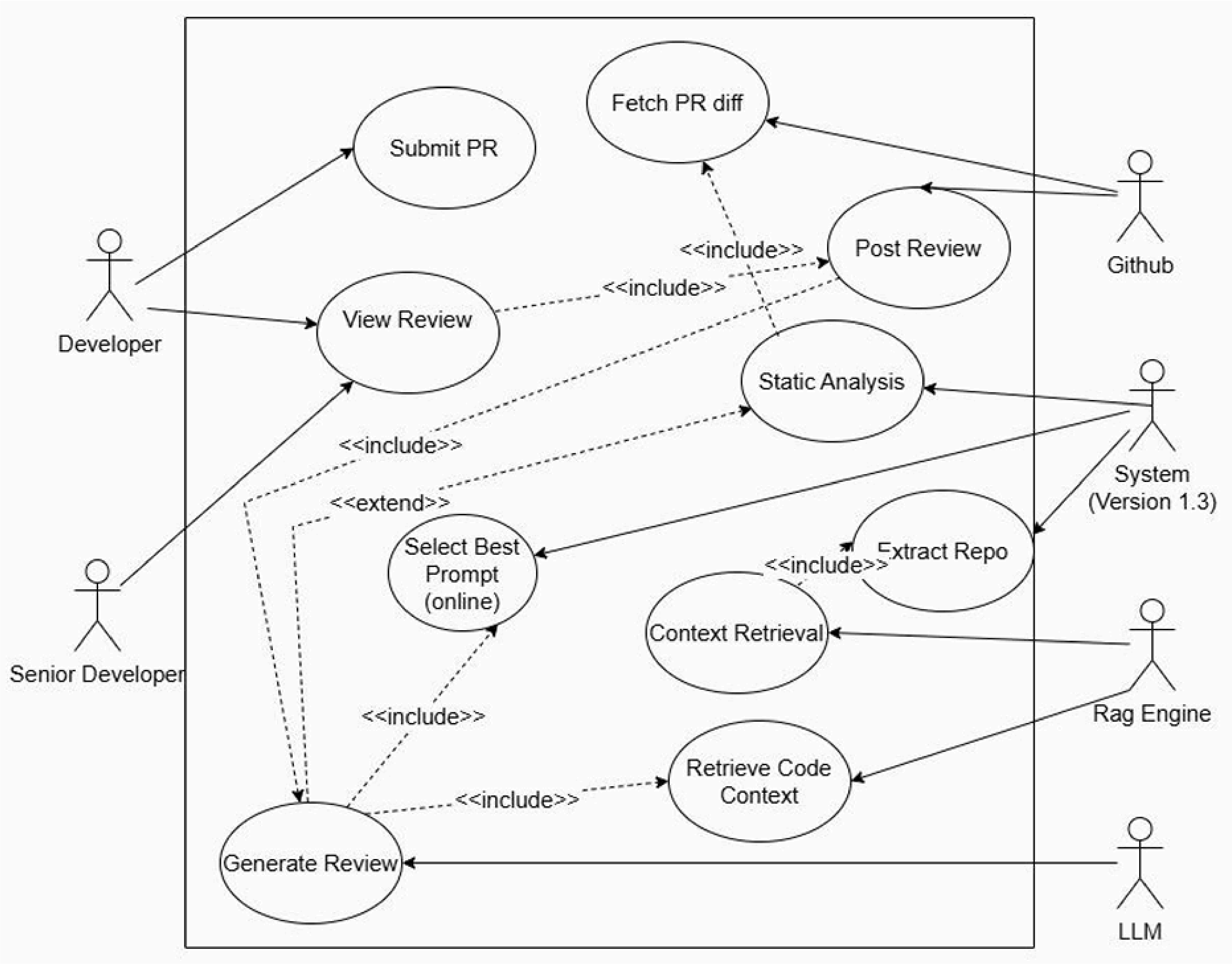
Sprint 4 :
Advanced Optimization + Fine-Tuning

Upgraded prompt selector to online-learning SGD, migrated RAG to Pinecone, and unified RAG + fine-tuned model into an optimized pipeline.

Sprint 5 :
Dashboard + Documentation + Testing

Added PR-history dashboard, completed all documentation, and performed full testing (unit, integration, GUI, load, spike).

Use Case Diagram



Takeaways

- AI alone is not reliable—grounding it with static analyzers significantly reduces hallucinations and increases trustworthiness.
- RAG boosts accuracy by using real repo context.
- GitHub workflow & webhook understanding is essential for smooth automation.
- Sprint-based development helps clarify scope and reduce confusion.
- Clear user stories make PR review workflow easy to design.
- LoRA models can be accurate but require heavy fine-tuning; large LLMs work better by default.
- Separating system components reduces complexity.

Achievements

- Built a fully automated PR-review pipeline using cloud LLMs integrated with GitHub Actions.
- Added Semgrep-based static analysis and RAG-based context retrieval for accurate, grounded reviews.
- Implemented full-repo indexing, adaptive prompt selection, and scalable retrieval via Pinecone.
- Developed a PR-history dashboard with insights, trends, and improved documentation/testing.
- Optimized CI runtime by modularizing components and switching to lightweight analyzers.
- Ensured secure credential handling and relied on GitHub-native logs for transparency.
- Delivered a production-ready AI code reviewer with inline comments and clear fallback messages on tool failures.

Mistakes

- Overestimated sprint capacity and planned an unrealistic online-to-local development workflow.
- Initially used heavy static analyzers (Pylint, Bandit, Flake8), not realizing they would slow GitHub workflows due to dependency installation.
- Switched late to Semgrep, which was more suitable and much faster for CI, causing avoidable rework.
- Underestimated integration complexity across GitHub Actions, static analysis, and RAG, leading to delays and mid-project adjustments.

Rating of Project Artefacts

• Requiremnet specifications	3.5/5	• RAG integration	4/5
• System design and architecture	3.5/5	• Dashboard	4/5
• Sprints & backlog	3.5/5	• Documentation	4.5/5
• ML model implemntation	4.5/5	• Project management	4/5

Overall Ratings

Pull panda – 8.5/10

Overall, our project demonstrates a strong balance of technical depth and practical implementation. Across all sprints, the team progressed from a simple cloud-based PR review pipeline to a fully context-aware RAG system with static analysis integration, reliable CI automation, and a functional dashboard. With complete documentation, stable workflows, and thorough testing, the system meets its goals effectively while leaving room for future enhancements.

We rate our project 8.5/10, with marks slightly deducted because—despite completing all major testing components—we were unable to finish mutation testing within the given timeline.

Team

Leader - Het Rank (202301081)

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Thank you!